SAMIKA GANDHI

https://samikagandhiportfolio.vercel.app/ . samikagandhi44@gmail.com . (346)668-7074 . linkedin.com/in/samika-gandhi

EDUCATION

B.S. BIOMEDICAL ENGINEERING, TEXAS A&M UNIVERSITY *(GPA - 3.5/4.0)* *AUG '23 – MAY '27*

Awards – Dean's Honor Roll, Aggie Research Scholar, Aggie Ring Scholarship, Jeff Summers BME Scholarship

PROFESSIONAL EXPERIENCES

LIFECYCLE ENGINEERING INTERN, JOHNSON & JOHNSON MEDTECH, CINCINNATI, OH *MAY – AUG '26*

- Developed and executed 300-trigger performance trials on porcine tissue for a tissue sealer to investigate customer complaint trends, conducting Cp/Cpk analysis on Minitab, eliminating trigger mechanism as root cause
- Integrated quality system for an acquired PMA-approved esophageal implant by overhauling FMEAs to align with J&J quality standards, strengthening risk management for a ~$300M product portfolio
- Designed a novel solution to enhance surgical robot performance after identifying a usability gap, presenting solution to Patent Review Board, potentially reducing operational time for surgeons by > 20%
- Facilitated an internal audit across 3 labs, verifying labeling, compliance records, personnel training and equipment IQ/OQ/PQ documentation to close compliance gaps and prevent recurrence

QUALITY ASSURANCE ENGINEERING INTERN, JOHNSON & JOHNSON MEDTECH, IRVINE, CA *MAY – AUG '25*

- Resolved nonconformance by developing CAPA and implementing sterilization verification controls, mitigating ~$60,000 in losses per work order
- Spearheaded root cause investigation for NPD catheter tip cracking through Fishbone diagram, identifying defect in supplier's measurement system through Variable Gage R&R, reducing failures by ~33%
- Engineered DOE-driven optimization of manufacturing qualification test scripts spanning 80 pages, simplifying test scripts and reducing operator error by 18%
- Developed a Kanban-based NPD component traceability system spanning R&D labs, supply chain inventory and packaging operations, cutting duplicate component orders by 25% per month, saving ~$70K/year

QUALITY SYSTEMS CO-OP, JOHNSON & JOHNSON MEDTECH, CINCINNATI, OH *JAN – MAY '25*

- Automated non-product software (NPS) ownership workflows using Power Automate, eliminating 260 hours of manual effort and ensuring compliance with ISO 13485
- Authored new NPS design inputs, unit-test strategies, acceptance criteria, and contingency plans, reducing anticipated system downtime during NPS failure by ~25%

PROJECTS

R&D ENGINEER (CAPSTONE PROJECT), ROCHE DIAGNOSTICS, COLLEGE STATION, TX *AUG '26 – MAY '27*

- Designing and developing a syringe-compatible device to administer high-viscosity ocular formulations, applying mechanical design principles to address high-force delivery challenges
- Defining and executing design verification and validation protocols and DFMEA, assessing device performance against user and design requirements, identifying and mitigating 200+ failure modes

RESEARCH ASSISTANT, PEDIATRIC DEVICE LAB (PI – DR. HARIDAS), COLLEGE STATION, TX *AUG '26 - PRESENT*

- Designing and integrating mechanical subassemblies for a fetal cardiac testbed that replicates cardiac flow and pressure conditions associated with HLHS, enabling V&V tests for a novel surgical tool for 3 different surgeries

DESIGN LEAD + 1ST PLACE, SLING HEALTH TAMU, COLLEGE STATION, TX *AUG '23 – APRIL '24*

- Led concept development for an IUD insertion training kit, translating user needs into geometry specifications
- Established design constraints through customer discovery with physicians, professors and nurses, conducting competitive analysis using a Pugh matrix across criteria like reliability and clinical relevance

SKILLS

- ***Prototyping*** – SolidWorks, AutoCAD, 3D printing, Technical drawing, GD&T, Rapid prototyping, Fixture design, DFM/DFA, Tolerance analysis, Poka-yoke
- ***Data Analysis*** **–** Python, Tableau, LabVIEW, SPC, KPI analysis, MS Excel, Minitab, Power BI, Technical writing
- ***Six Sigma Principles*** – Gage R&R, DOE, DMAIC, Verification & Validation, FMEA, IQ/OQ/PQ, 5S, Value Stream Mapping, Process Capability Analysis